[CE310] Evolutionary Computation and Genetic Programming

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Evolution strategies

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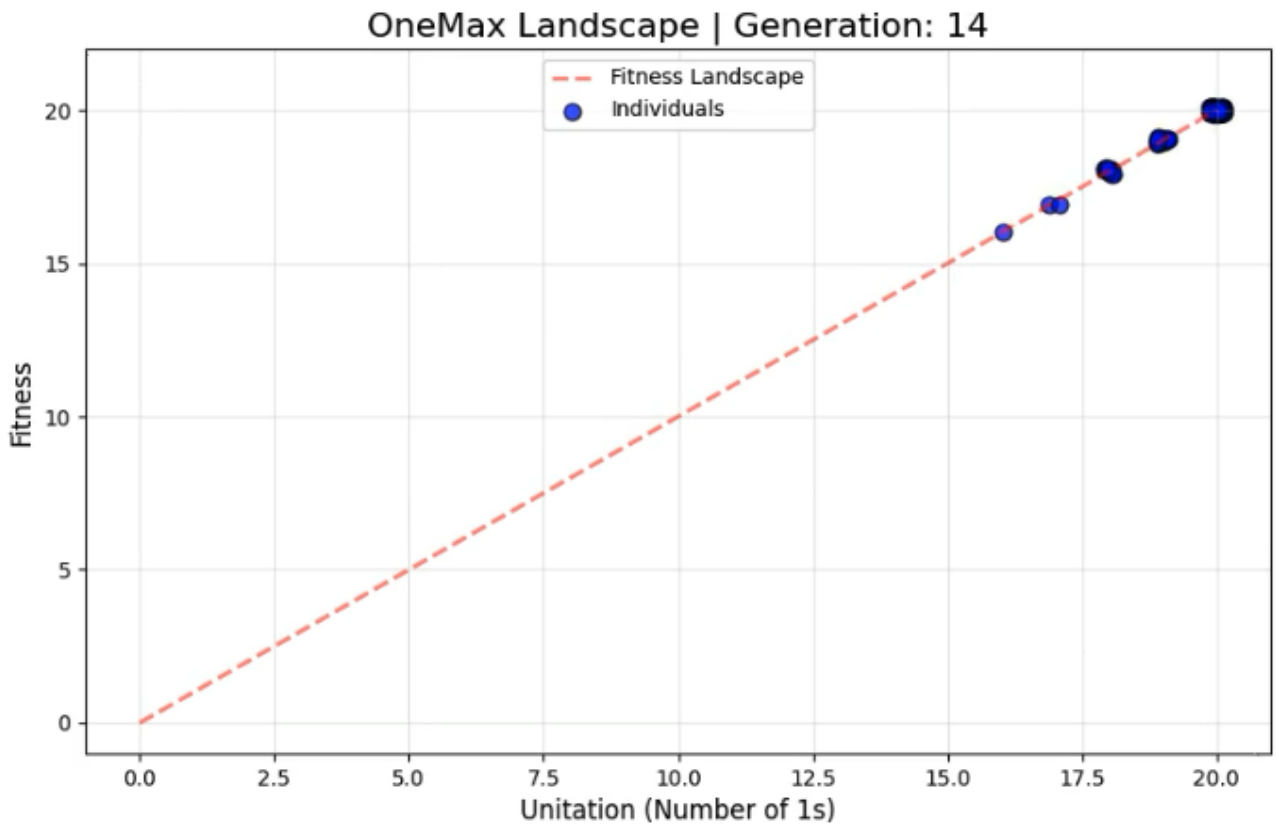
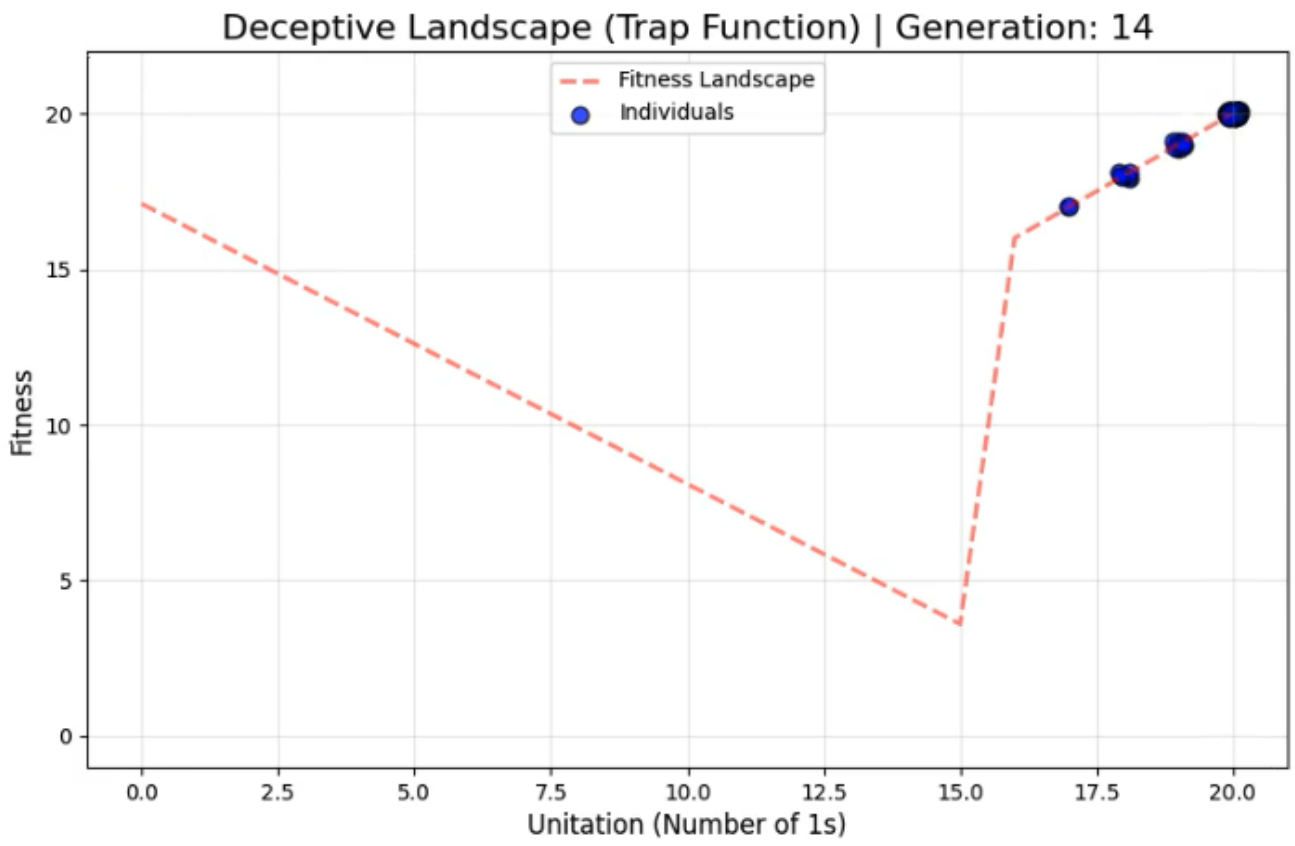
Post Lab Addendum

Binary Steady State GA on OneMax function



Binary Steady State GA on deceptive/trap function



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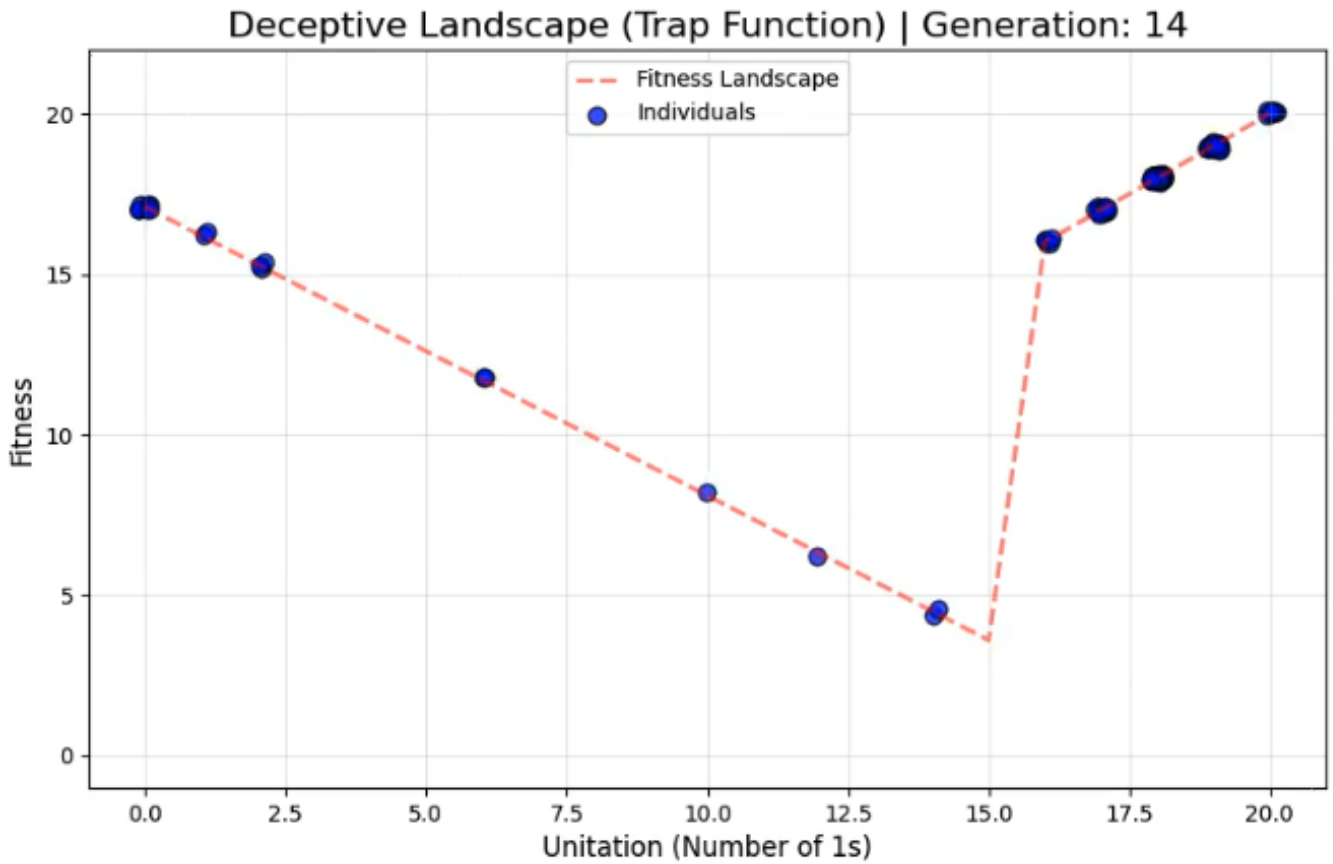

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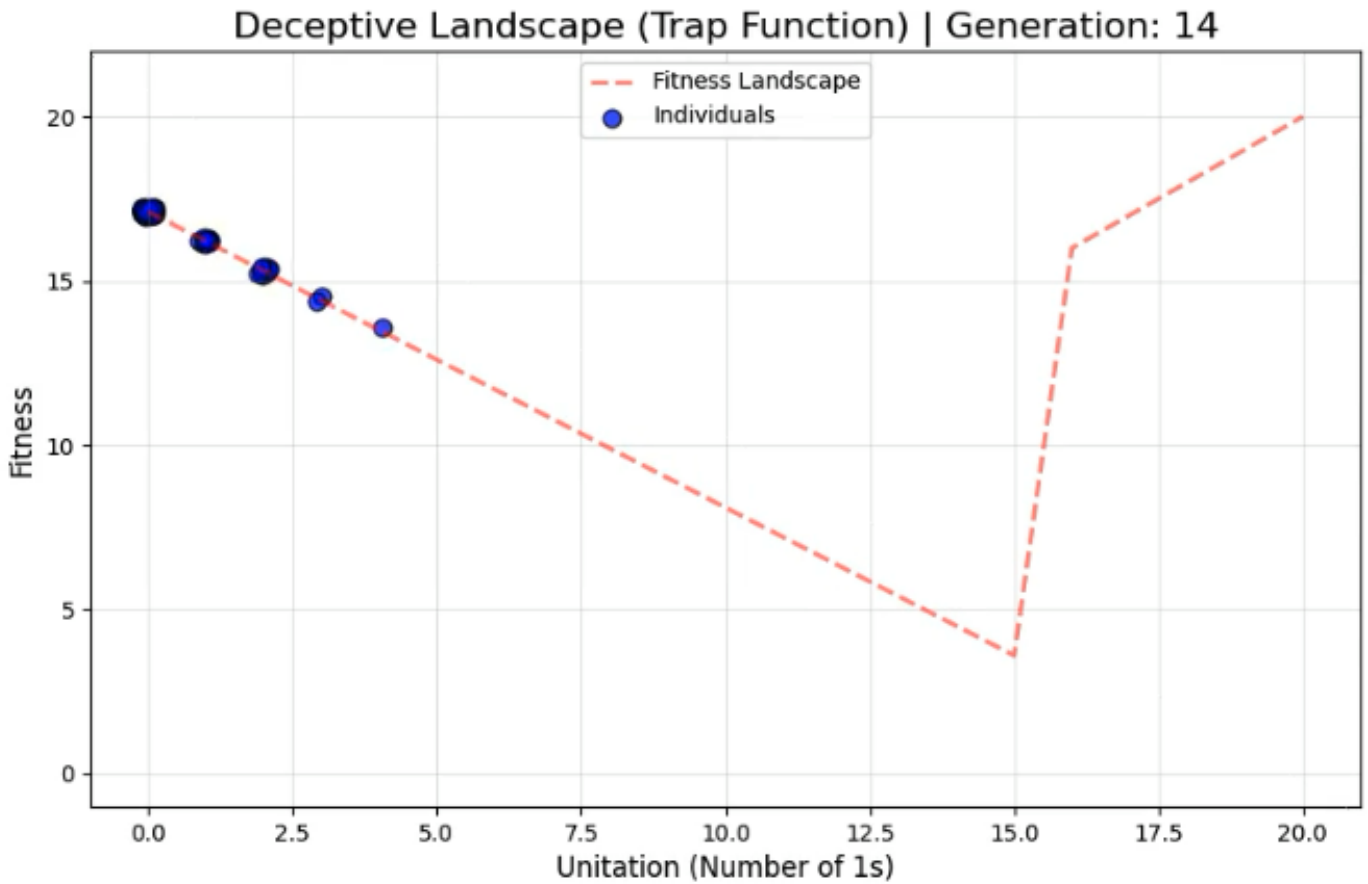


Today’s learning objective

- Describe **evolution strategies** (ESs) algorithm and possible ways to implement them.
- Discuss and contrast **evolution strategies** (EAs) and **genetic algorithms** (GAs)

Evolution Strategies

- An **Evolution Strategy** (ES) is EAs which uses arrays of real numbers as a representation, like real-valued GAs.
- **Mutation** is the key variation operator (unlike GAs, where recombination is usually deemed to play that role)
- **Mutation** = **addition of Gaussian random numbers** to the floats that represent the individuals being optimised.
- The **amplitude of deviates is varied dynamically** to maximise performance (but often ESs are also used with fixed mutation).

Gaussian Mutation

- Same implementation as mutation in real-valued GAs:

$$\text{Offspring}_{Param1} = \text{Parent}_{Param1} + \text{sigma} \cdot R_1$$

$$\ldots$$

$$\text{Offspring}_{ParamN} = \text{Parent}_{ParamN} + \text{sigma} \cdot R_N$$

but

$R_1, \ldots, R_N$ = random numbers drawn from a **standardised Gaussian distribution**

Gaussian Mutation

Same implementation as mutation in real-valued GAs but the R_i are NOT uniformly distributed

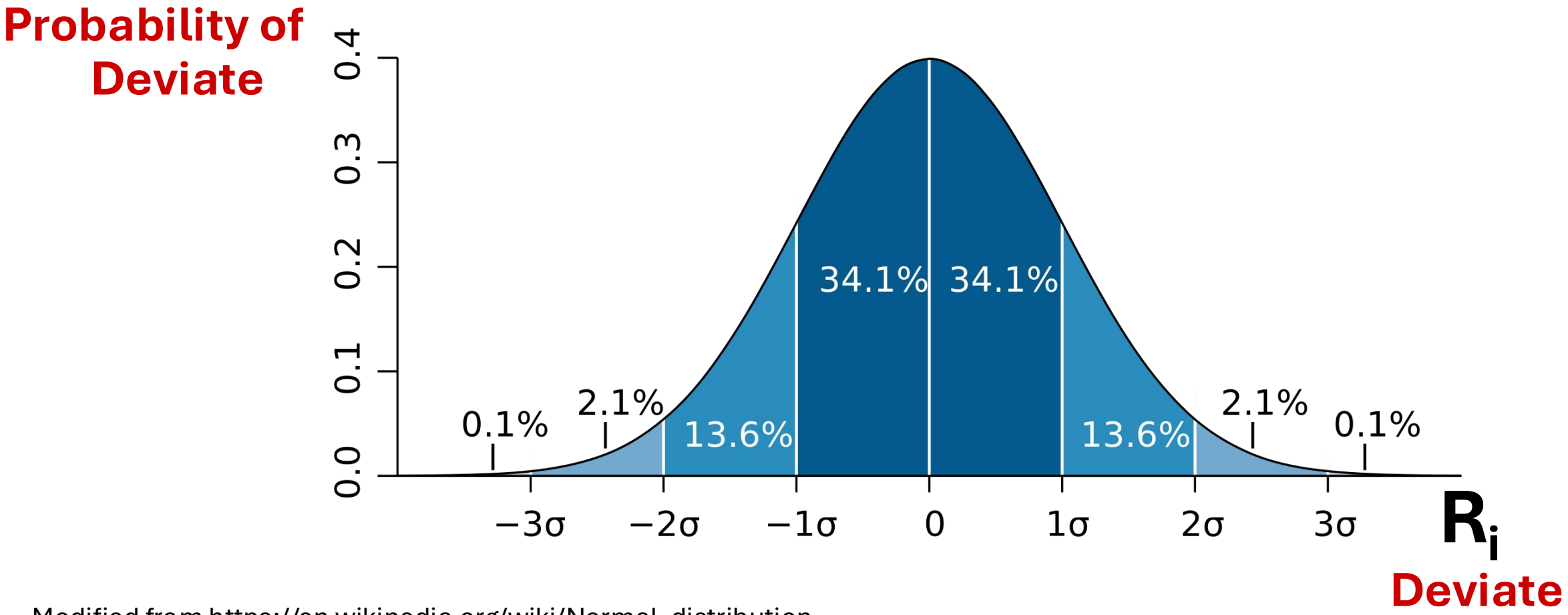
$$\begin{aligned} \text{Offspring}[1] &= \text{Parent}[1] + \text{sigma} \cdot \mathrm{R}_1 \\ &\vdots \\ \text{Offspring}[N] &= \text{Parent}[\mathrm{N}] + \text{sigma} \cdot \mathrm{R}_N \end{aligned}$$

where:

- *sigma* is a suitable positive constant and
- $\mathrm{R}_1, \ldots, \mathrm{R}_\mathrm{N}$ are (different) random numbers drawn from a <u>normal/gaussian</u> distribution
- In python:

```
import random
Ri = random.gauss(0, sigma)
```

Standardised Gaussian Deviates

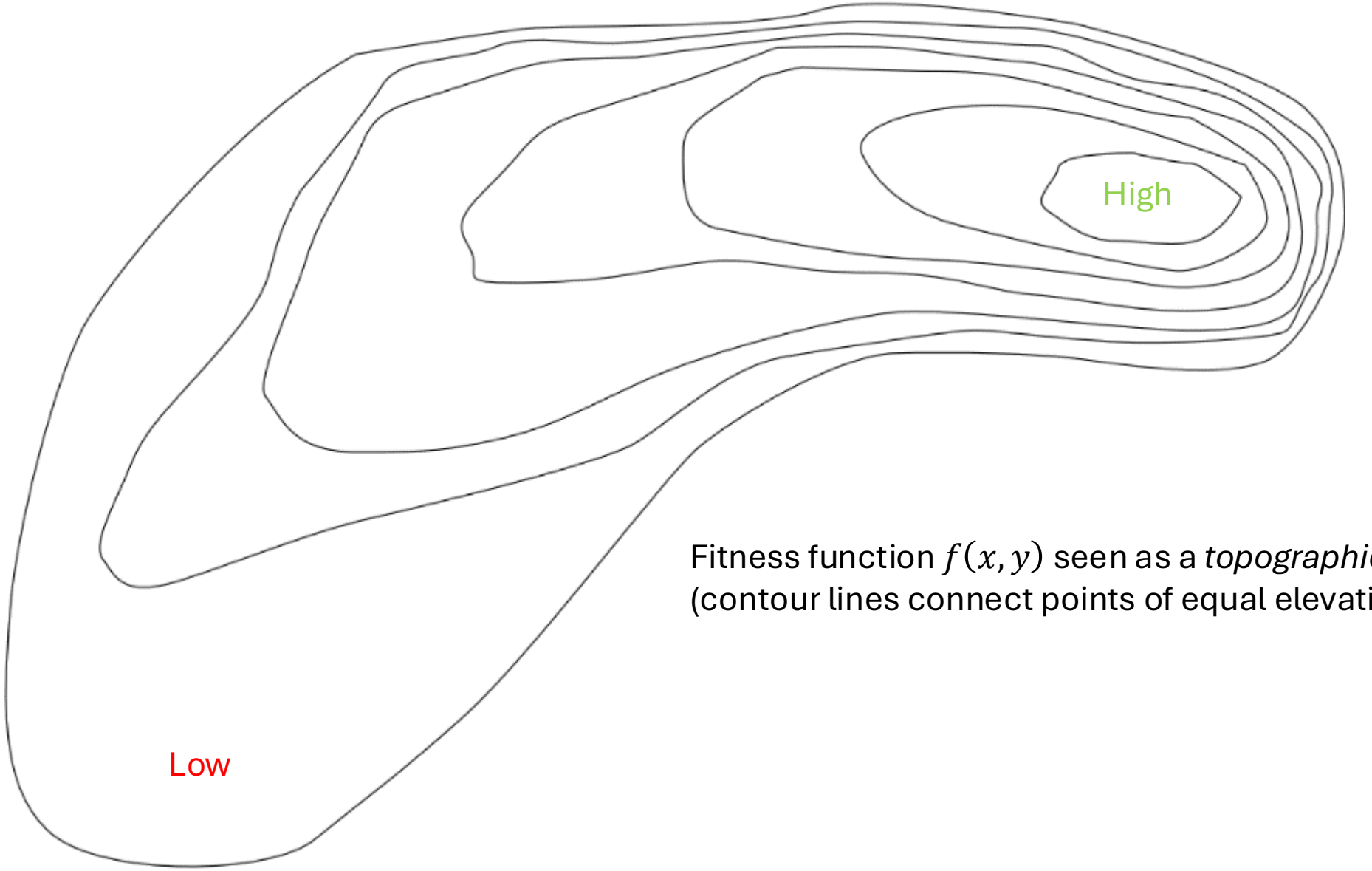


Modified from https://en.wikipedia.org/wiki/Normal_distribution

(1+1) ES

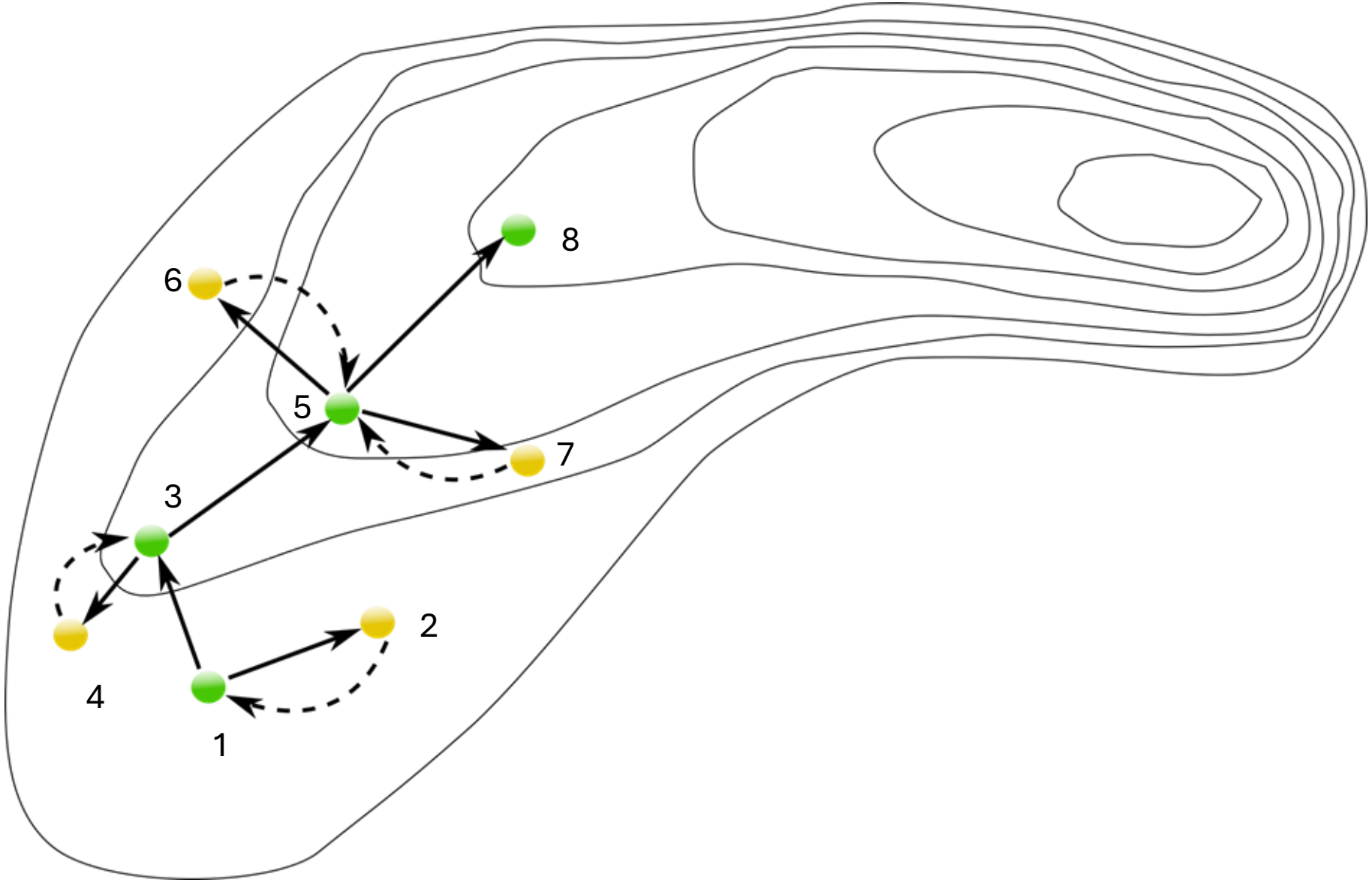
- The (**1+1**) **ES** uses a population consisting of only one individual (the first “1” in “(**1**+1)”).
- The parent generates one offspring (the second “1” in “(1+**1**)”) per generation by applying Gaussian mutations.
- If offspring performs no worse than its parent, it replaces it.

Sample 2-D Fitness Landscape



Fitness function $f(x, y)$ seen as a *topographic map* (contour lines connect points of equal elevation)

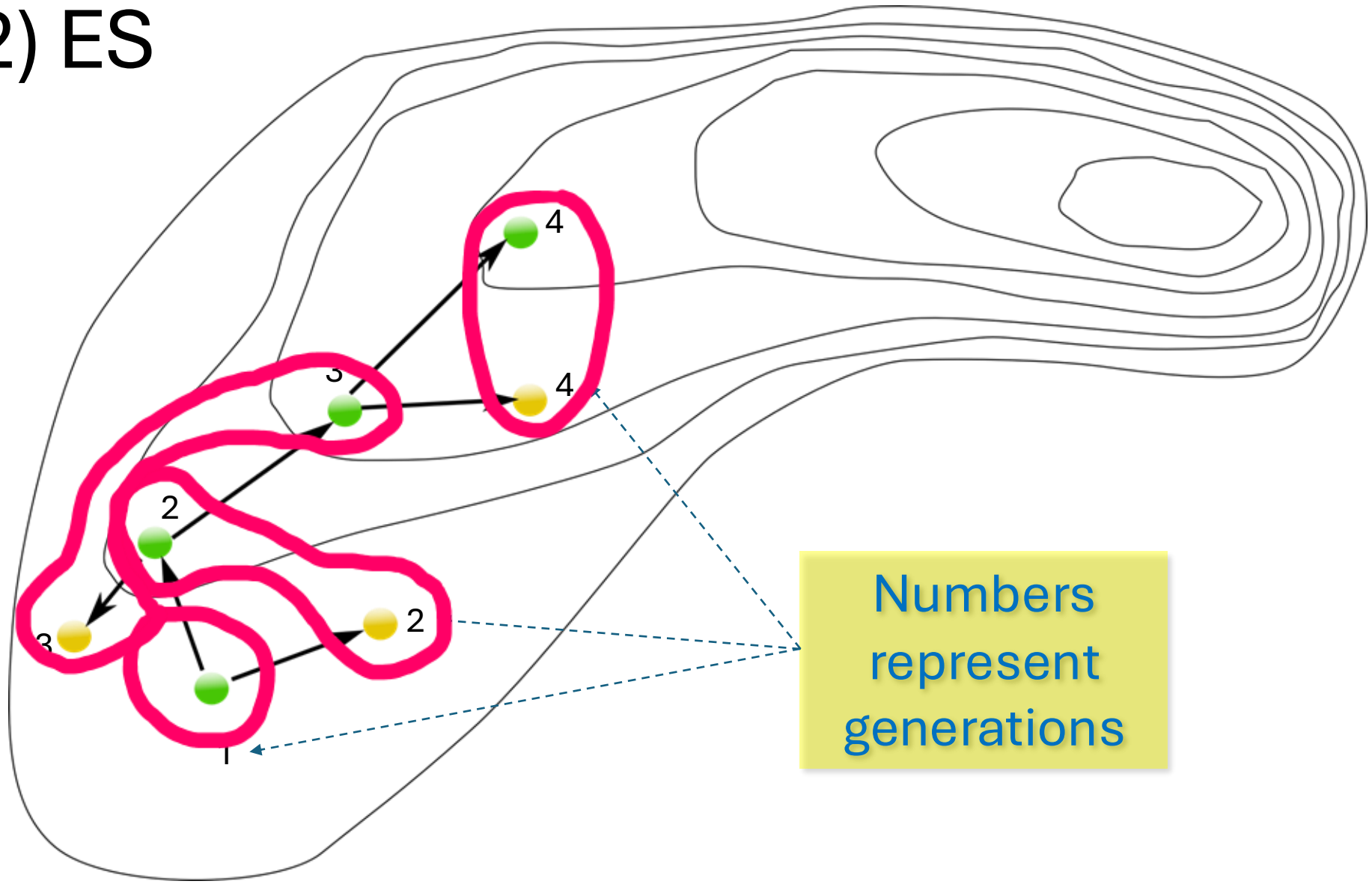
(1+1)-ES Behaviour



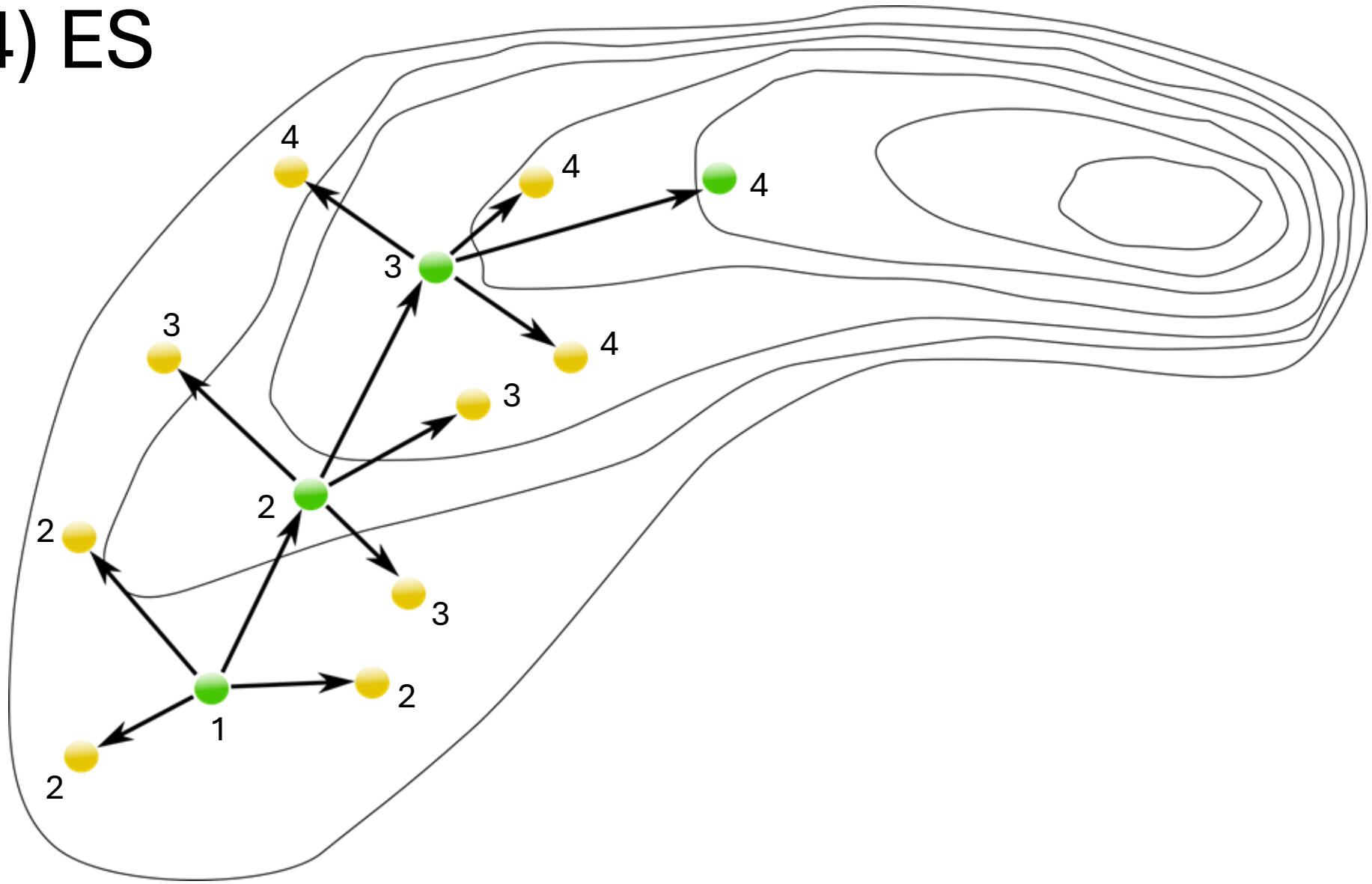
(1+lambda), (mu+lambda) and (mu,lambda) ES

- **(1+lambda**) ES is like the (1+1) ES except it generates **lambda offspring** and it chooses the best as a candidate for replacing the parent.
- **(mu+lambda)** ES produces **lambda offspring** at each generation **via crossover and mutation**. The best mu individuals out of the lambda + mu available ones are retained.
- **(mu, lambda)** ES is like (mu+lambda) ES but it the **replaces the parents with the best mu offspring out of lambda generated** (naturally, lambda > mu for this to work).

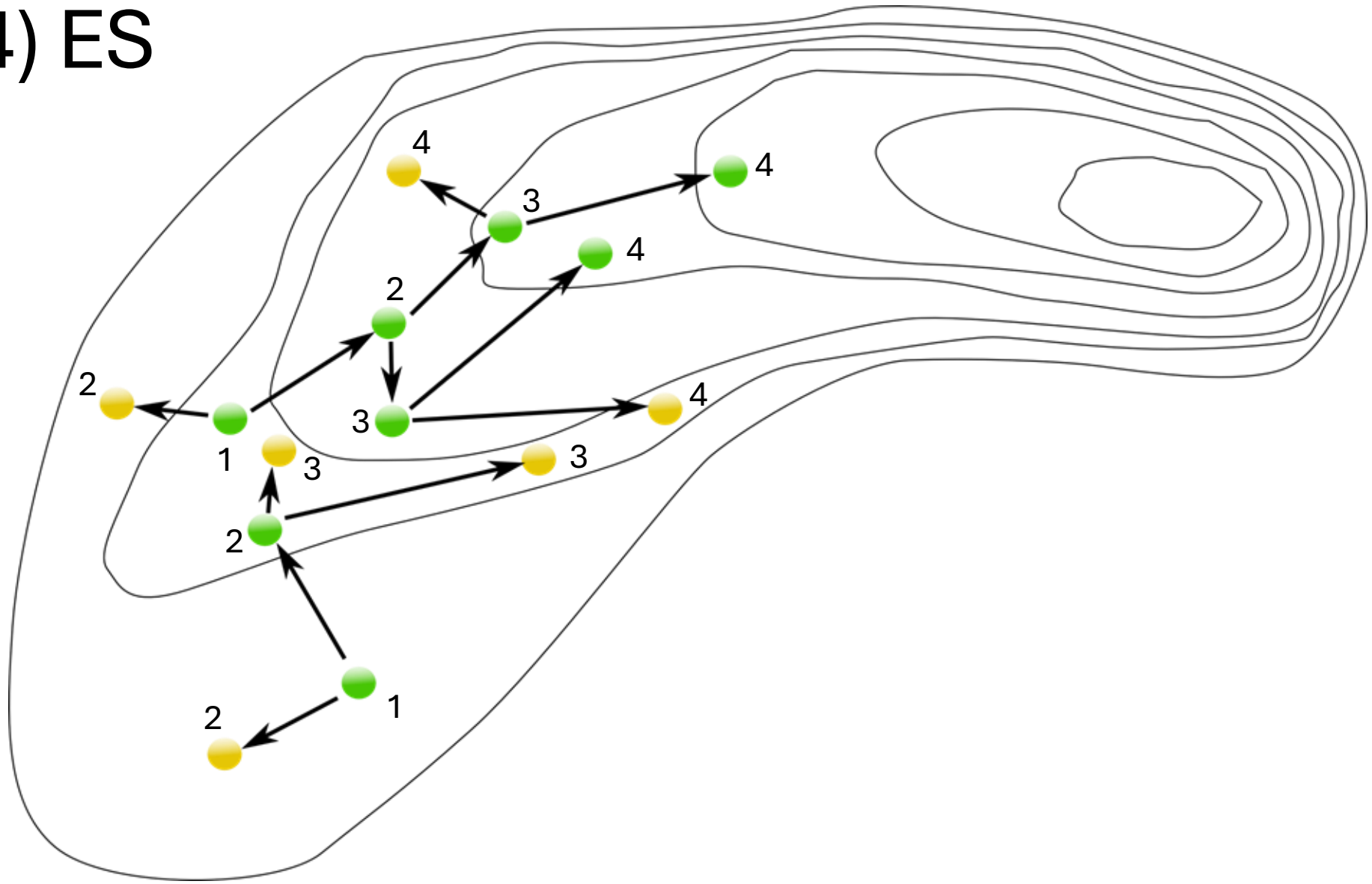
(1,2) ES



(1,4) ES



(2,4) ES



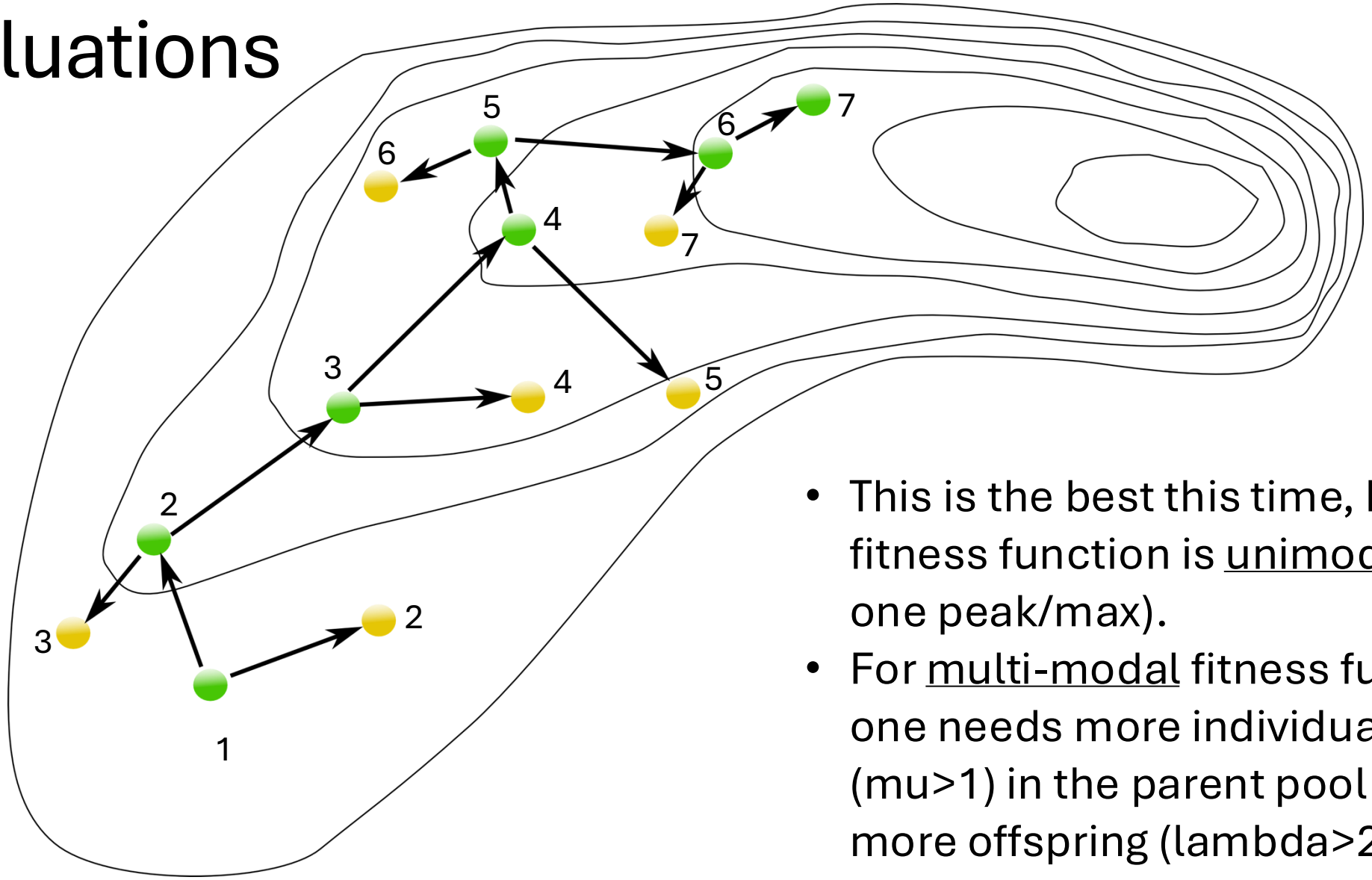
Which ES is best?

- It might seem that (1,4) and (2,4) ES are **best**.
- However, more offspring means more fitness evaluations, i.e., longer runtimes

Our example ES	Fitness evaluations
1+1	8
1,2	7
1,4	13
2,4	14

- So, fair comparison should be done on approximately same number of fitness evaluations

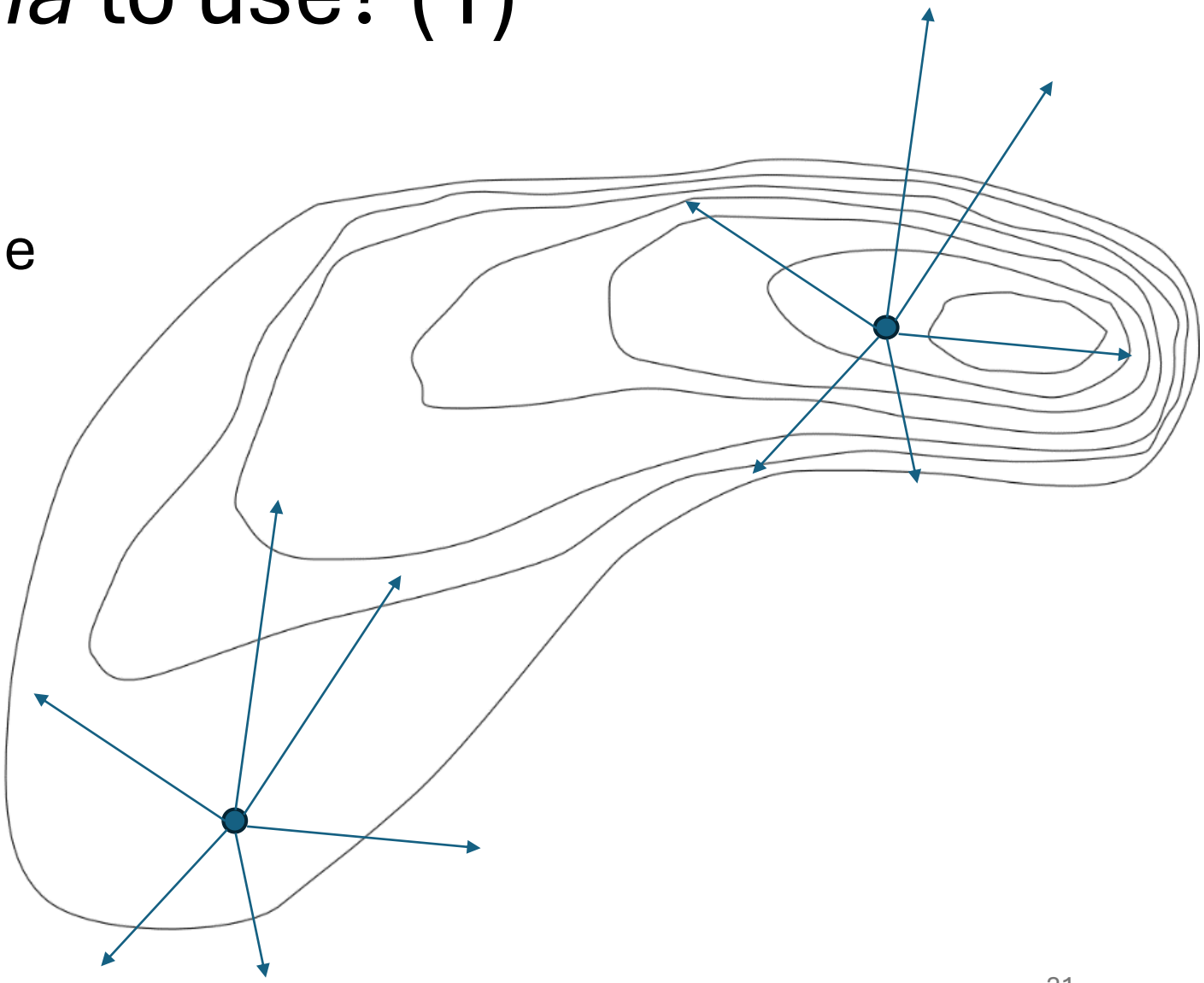
(1,2) ES run extended to 13 fitness evaluations



- This is the best this time, but the fitness function is <u>unimodal</u> (only one peak/max).
- For <u>multi-modal</u> fitness functions one needs more individuals (mu>1) in the parent pool and more offspring (lambda>2)

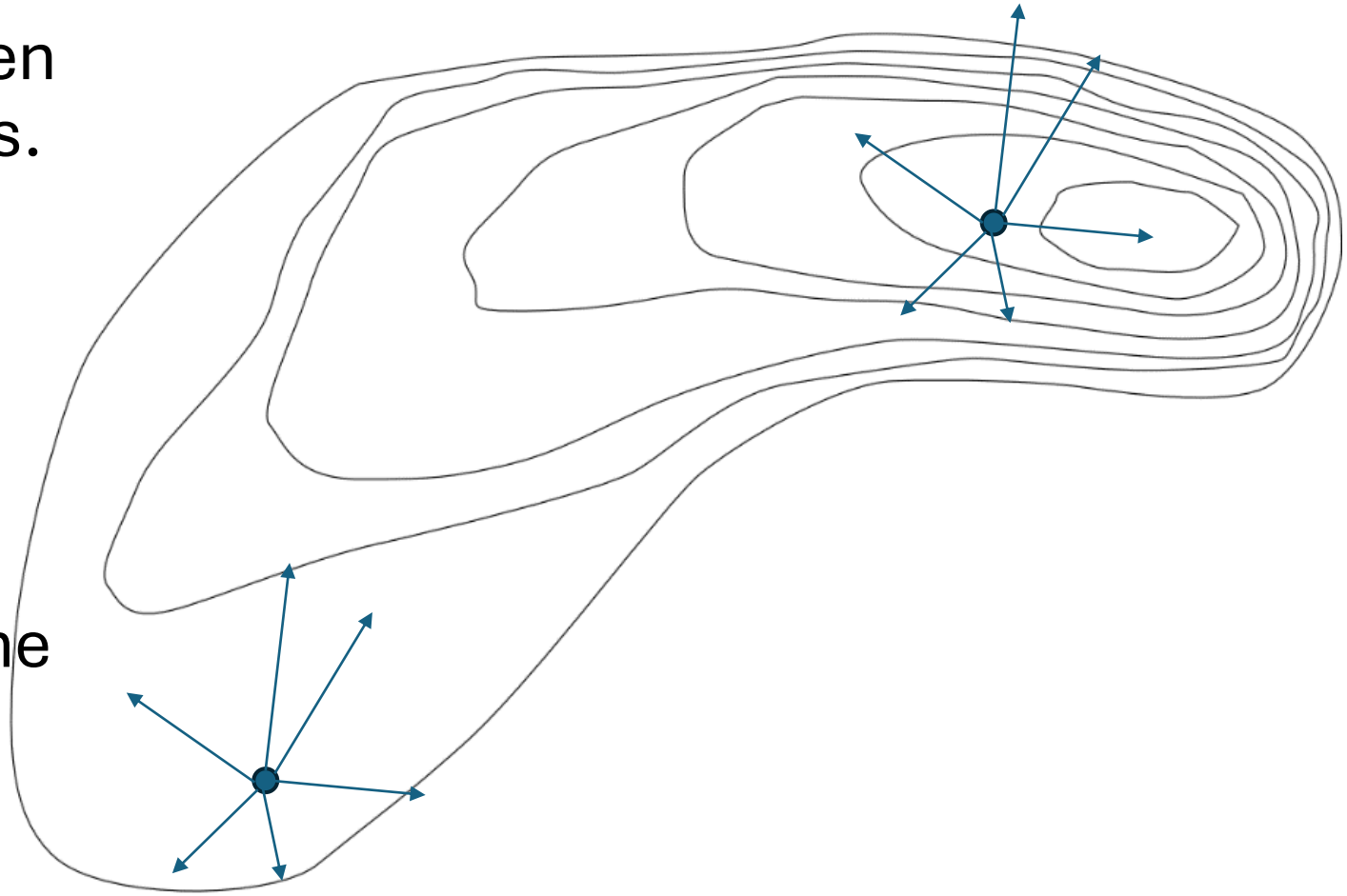
Which value of *sigma* to use? (1)

- **Big** *sigma* means children will be further away from the parents.
- Initially this may lead to bigger improvements in fitness
- However, the closer the population gets to the optimum the harder it gets to home on it.



Which value of *sigma* to use? (2)

- **Small** *sigma* means children will be closer to the parents.
- Initially this may lead to smaller improvements in fitness
- However, the closer the population gets to the optimum it is easier to home on it.



Sigma Adaptation and 1/5 Rule

- This issue is addressed by the **1/5 rule,** where **sigma** is changed according to the frequency of successful mutations
- We define

$$s_{mut} = \underline{\text{fraction of times children are fitter than parents}}$$

- If $s_{mut} > 1/5$ ➔ increase sigma

  # we're exploring too little, e.g., on a gentle slope
- If $s_{mut} < 1/5$ ➔ decrease sigma

  # we're exploring too widely, e.g., near the optimum
- If $s_{mut} = 1/5$ ➔ do nothing # we're exploring just right!

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Next week:
Genetic Programming

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